$$\frac{Dp}{Dt} = \frac{\partial p}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial w}{\partial z}$$

$$\int \nabla \cdot G = \oint_c G \cdot n \, dA$$

$$\frac{1}{r}\frac{\partial r \rho u_r}{\partial r} + \frac{1}{r}\frac{\partial \rho u_\theta}{\partial \theta} + \frac{\partial \rho u_z}{\partial z} = 0$$

(13) Continuity (12) 2nd law Conservation

Angular momentum Principal equation

$F = Iw$

(19) Volumetric Strain slow both

(18) Bernoulli turbine

$u = 2.5 + 0.9x^2$

(17) $\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0$

(16) $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

$\frac{\partial u}{\partial x} = 1.8x \quad \frac{\partial u}{\partial z} = 0$

(14) $\frac{\partial v}{\partial y} = ?$

(15) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} = 0$

$\frac{\partial v}{\partial y} = 1.8x$

$1.8x + \frac{\partial v}{\partial y} = 0$

$v = 1.8xy + f(x)$

Navir (30) $-\nabla p + \rho g$

Streamline (29) $\rightarrow q$

Stra (28) $\frac{\partial w}{\partial y}$ u=x

0 27 0.8-0.8

layer 26 P-P∞

viscos 25 B-layer

24 newton

$u_r = 0 \; u_\theta = 0 \; \text{All}$ — 23 Luminar

$\frac{\partial u}{\partial t} = 0 \quad 2\mu\frac{\partial u}{\partial x}$ 22 Tax

21 nonlinear

$\mu\nabla^2 v$

20 — missing

Continue

13 12 11 10 9 8 7 6 5 4 3 2 1

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

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Steady Incom

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$P + \frac{1}{2}\rho V^2 + \rho g h = \text{Constant}$$

$$P = \frac{F}{A} = m$$

(39) $u = 2.5y + 0.9x^2$ ABS $= 1.8x$ or $1.8xy + Ry$ (5)

(38) Incomp Homogenous (5)

(37) Direct Faster (5)

(36) Compressible cylindrical coordinate (3)

$$\frac{1}{r}\frac{\partial r \rho u_r}{\partial r} + \frac{1}{r}\frac{\partial \rho u_\theta}{\partial \theta} + \frac{\partial \rho u_z}{\partial z} = 0$$

(35) $\int \nabla \cdot G \, dv = \int_{\text{Surface}} G \cdot n \, dA$ (3)

(34) $V = r\omega$ (3)

(33) $\frac{Dp}{Dt} = \frac{\partial p}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial w}{\partial z}$ (2)

(32) Kinematic Similarity (2)

(31) Newton 2nd law (2)

(11) Match Number $= \frac{V}{\frac{\text{Sound}}{\text{Speed}}} = M$

(10) Kinematic Dynamic Same

Vgeo not Kinq

Gpes – dynamic

Dyna – Kinematic

(9) angle of Rotation

(8) each term with Pressor Bernoulli

(7) Primary → (7)

(6) Vector Product

(5) $C = 2\pi r$

(4) Volume not eq Sol

(3) Linear $= mv$ mom

(2) X

(1) atmo / gage

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(Math 642) (MCQ'S)

1- The dimension of Froud no is
(a) m/L^2 (b) mass x length (c) {1} (d) 0

2- We can ~~de~~ make $F=ma$ nondimensionalized by dividing with
(a) Force (b) mass (c) acceleration (d) arbitrary const

3- Dimension of speed is
(a) $\frac{L}{t}$ (b) . . .

4- The stress tensor σ_{ij} contains components ____.
(a) 6 (b) 9 (c) 3

5- The continuity equation is used to ____.
(a) used to find missing component
(b) none (c) both (d) to find volume . . .

6- Which statement is true ____.
(a) The continuity equ used to energy
(b) energy both (c) none (d) Incompressibili

7- The pressure close to boundary layer
(a) $P \approx P_0$ (b) $P - P_0 \simeq 0$ (c) Yad nahin

8- The diagonal elements in the σ_{ij} are stress

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M T W T F S DATE:

(a) strain rate (b) stress (c) ?

9- The Bernoulli equ holds for fluid
(a) steady, incompressible, irrotational
(b) steady, compressible, rotational
& vice versa.

10- law of conservation of mass is related to ____.
(a) Continuity equ (b) Yad nature.

11- In Navier stock equ which term is non linear
(a) $-\nabla P$ (b) $\mu \nabla^2 \vec{V}$ (c) ρg

12- The stream function holds
(a) $u = \frac{\partial \psi}{\partial y}$ & $v = -\frac{\partial \psi}{\partial x}$

13- If $u = 0.5 + 0.8x$ & $v = 1.5 - 0.8y$ then it is:
(a) incompressible (b) compressible
(c) ?

14- The incompressible two dimensional continuity equ in yz plane is
(a) $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

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(1)

Two Dimentional compressible flow

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0.$$

Compressible cylindrical coordinates

$$\frac{1}{r}\frac{\partial(r\rho u_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho u_\theta)}{\partial \theta} + \frac{\partial(\rho u_z)}{\partial z} = 0$$

Compressible flow continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho V) = 0$$

Compressibl three dimensional flow:-

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0.$$

(2)

Two Dimensional incompressible flow

$$\partial u/\partial x + \partial v/\partial y = 0.$$

Incompressible ~~dij~~ cylindical coordinates:-

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{1}{r}\frac{\partial(u_\theta)}{\partial \theta} + \frac{\partial(u_z)}{\partial z} = 0.$$

Incompresible flow continuity equation:

$$\nabla \cdot V = 0$$

Incompressible three dimensional flow

$$\frac{\partial \rho}{\partial t} = 0$$

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MTH 642 8:00AM

Quiz

① Bernouli eqn will not be valid
① if flow is steady and frictional effects negligible
② " " " incompressible and " " " "
③ " " " " and compressible
④ None

② The diagonal component of stress tensor a_{ij} are named as
(i) viscous stress (ii) sneaar stress (iii) normal stress
(iv) turbulent stress

③ Cont eqn in cartesian -- for incompressible two dimen flow in YZ-plane?
(i) $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$ (ii) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ (iii) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$
(iv) $\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0$

④ $\vec{\nabla} \cdot \vec{v} = 0$ is named as continuity eqn for
(i) unsteady incompressible flow (ii) Steady compress
(iii) steady incompress

⑤ Reynold theorem is define as ——
(i) $\{1\}$ (ii) $\{0\}$ (iii)
time rate mile

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Quiz MTH 642 8:00 AM

① Bernouli eqn will not be valid
il flow is steady and frictional effects negligible

14-March-2022 MTH 642 Paper Solve ZAIN AHSAN

Q What is rotational ~~motion~~ motion and quantities? (2)

Ans A motion in which during which all Points in the body move in circle about the axis of rotation. It also described angular quantities such as θ, $\vec{\omega}$ and $\vec{\alpha}$

Q Condition of model and Prototype? (2)

Ans First Condition is geometric Similarity and 2nd condition is Kinematic Similarity.

Q expression $\vec{\nabla} \cdot (\rho \vec{V})$? (2)

Ans Steady Compressible flow Continuity eqn

Q Angular momentum Principle in mathematical exp-? (3)

Ans $$\sum \vec{M} = \frac{d}{dt} \int_{CV} (\vec{r} \times \vec{v}) \rho dV + \sum_{out} (\vec{r} \times \dot{m}\vec{v}) - \sum_{in} (\vec{r} \times \dot{m}\vec{v})$$

when there is no external force the Angular mo---

$$= \frac{d\vec{H}_{CV}}{dt} + \sum_{out} (\vec{r} \times \dot{m}\vec{v}) - \sum_{in} (\vec{r} \times \dot{m}\vec{v})$$

Q Differiak b/w dimension and unit? (3)

Ans Dimension are physical quantities that can be measured whereas units are arbitrary names that correlate to particular dimension

example a dimension is length whereas a meter is unit (that describe length)

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MTH 642 MCQ's 2022

1) The dimension of froud no is
a) m/L^2 b) mass X length c) $\frac{L}{T^2}$ d) 0

2) We can make $F=ma$ non dimensionalized by dividing with
a) Force b) mass c) Acceleration d) Arbitrary Constant

3) Dimension of speed is L/t, Lt^{-1}

4) The stress tensor σ_{ij} contains components
a) 6 b) 9 c) 3 (Topic 135, Pg 145)

5) The continuity equation is used to
a) Find missing component b) none
c) both d) To Find Volume

6) Which statement is true
a) The continuity eq used to energy b) both
c) none d) Incompressibility.

7) The Pressure close to boundary layer
$\rho - \rho_0 = 0$

8) The diagonal elements in the σ_{ij} are
a) Strain rate b) stress

9) The Bernoulli eq holds for fluid :
Steady, incompressible, irrotational

10) law of conservation of mass is related to Continuity eq

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11) The Navier Stocll eq which term is non linear $\mu(\nabla^2\vec{V})$

12) The stream function hold $V_x \frac{\partial \psi}{\partial y}$ & $V_z = \frac{-\partial \psi}{\partial x}$

13) If $V_x = 0.5 + 0.8x$ & $V_z = 1.5 - 0.8y$ then It is incompressible

14) The incompressible two dimensional continuity eq in yz plane is $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

15) The incompressible ~~two~~ dimensional xz continuity eq is $\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0$

16) Circumference of circle is $2\pi r$ b) πr^2 c) πr

17) In dimensional analysis is ~~not~~ true Geometric Similarity is necessary ~~for~~ & ~~Sufficient~~ Yet not have Kinematic ~~for dynamics~~ Similarity.

MTH642 Subjective

(Topic 07)

1) Write three Purpose of dimensional Analysis (3 Marks)

Ans:- 1) To generate non dimensional Parameters that helps in the design of experiments (Physical and/or numerical) and in the reporting of experimental results.

2) To obtain Scaling laws so that Prototype Performance can be Predicted from model Performance.

3) To (Sometimes) Predict tends in the relationship between Parameters.

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(Q2) Expand material Derivative $\frac{D\rho}{Dt}$ (2 marks)

Ans:- Expand equation

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) = 0$$

by using the Product rule on the divergence term. $\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \cdot \vec{V}) = \underbrace{\frac{d\rho}{dt}}_{\text{Material derivative of } \rho} + \vec{V} \cdot \vec{\nabla}\rho + \rho \vec{\nabla} \cdot \vec{V} = 0$

(Q3) What is the eq of law of conservation of mass. State the law (2 marks)

Ans:- Conservation of a mass for a CV:-

$$0 = \int_{cv} \frac{\partial \rho}{\partial t} dV + \int_{cv} \rho \vec{V} \cdot \vec{n} \, dA$$

$$\int_{cv} \frac{\partial \rho}{\partial t} dV = \sum_{in} \dot{m} - \sum_{out} \dot{m}$$

The net rate of change of mass within the control volume is equal to the rate at which mass flows into the control volume minus the rate at which mass flows out of the control volume.

(Q4) State linear momentum (2 marks)

Ans:- The Product of mass and the velocity of the body is called Linear momentum it is denoted by $P = mv$

16 march 2022

(Q5) Write Continuity eq for Compressible in cylindrical coordinates (3 marks).

Ans:- $\frac{1}{r} \frac{\partial (r v_r)}{\partial r} + \frac{1}{r} \frac{\partial (v_\theta)}{\partial \theta} + \frac{\partial (v_z)}{\partial z} = 0$

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MTH642 Quiz 16 March 2022

1) To which following flow the Bernoulli equation is applicable?
a) To a flow section that involves turbine
b) To a flow section past a machine
c) To a rotating flow in a tank.
d) To a flow section that involves impellers.

2) Which of the following represent the τ_{xx} component of the viscous stress tensor?
a) $\tau_{xx} = \mu\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$ b) $\tau_{xx} = \frac{\partial v}{\partial x}$ c) $\tau_{xx} = 2\mu\frac{\partial v}{\partial x}$
$\tau_{xx} = \mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$
(Topic 134, Pg 150)

3) Which of the following statement(s) are true about the continuity equation
a) Both b) None c) The continuity equation is used in determining the volumetric strain rate of steady incompressible flow.
d) The continuity equation is used in determining the incompressibility of an unsteady two dimensional flow.

4) The dimension of a dimensionless quantity is
a) {∞} b) {0} c) {1} d) {1/time} (Topic 101)

5) There are ____ components of a stress tensor
a) 9 b) 27 c) 3 d) 6

6) For steady, incompressible, laminar flow of a Newtonian fluid
a) $u_x = 0$ b) All of the given options c) $u_\theta = 0$ d) $\frac{\partial u_r}{\partial \theta} = 0$

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7) Which of the following is the order of magnitude of pressure within the boundary layer?

a) $P \sim P_\infty$ b) $P \sim \rho U^2$ c) $P_\infty \sim \rho U^2$

✓ d) $P - P_\infty \sim \rho U^2$

8) From which of the following how we deduce that a body tends to preserve its state of inertia?

a) Newtonian law of viscosity ✓ b) Newton's first law of motion c) Newton's second law of motion d) Newton's third law of motion.

9) In dimensional analysis

✓ a) Geometric similarity is a necessary condition for dynamic similarity b) Dynamic similarity is a necessary condition for Kinetic similarity.

✓ c) It is possible to have geometric similarity, yet not have Kinematic similarity.

d) Kinematic similarity is a necessary and sufficient condition for dynamic similarity.

10) Newtons second law of motion applied to a fluid particle in a flow is also named as a) law of conservation of mass

✓ b) law of conservation of energy

c) law of conservation of pressure

d) law of conservation of momentum

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11) Froude number is a ____ a ____ (Topic 106, Pg 110)
a) dimensionless Parameter b) dimensional Variable
c) dimensional constant d) dimensional group

12) Which of the following term of Navier-Stokes equation is nonlinear?
a) $(\vec{V}.\vec{\nabla})\vec{V}$ b) $-\vec{\nabla}P$ c) $\frac{\partial \vec{V}}{\partial t}$ d) $\mu\nabla\vec{V}$

13) The angular distance travel by any point in a rigid body during a complete rotation is
a) π rad b) $\pi/2$ rad c) $2/\pi$ rad d) 2π rad

14) Which of the following coordinate(s) is/are same in both the Cartesian coordinate and the cylindrical coordinates?
a) Z and y coordinates b) x and y coordinates (Topic 123, Pg 131)
c) Y coordinates d) Z coordinates

15) Which of the following is a definition of stream function? $u = \frac{\partial \psi}{\partial y}$ and $v = -\frac{\partial \psi}{\partial x}$ (Topic 131, Pg 138)
a)

16) If $u = 0.5 + 0.8x$ and $v = 0.5 - 0.8y$, then the flow field is a) compressible b) incompressible
c) both compressible and incompressible
d) not satisfy the continuity equation.

17) The continuity equation in Cartesian coordinates for incompressible, two dimension flow in the yz-plane will be given as (a) $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$
b) $\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0$ c) $\frac{\partial w}{\partial x} + \frac{\partial v}{\partial y} = 0$ d) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

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18) Divergence theorem is also called as Gauss's theorem (Topic 119 ; Pg 136)

19) Mach Number is defined as the ratio of flow speed of Sound. What would be the dimension of Mach Number b) {length/time}
a) {1} c) {0} d) {time/length}

20) Which of the following Statement(s) is/are true about the continuity equation?
a) The continuity equation is used in determining missing component of the velocity field of a steady two dimensional incompressible flow
b) Both c) None d) The continuity equation is used in determining the energy changes of an unsteady compressible flow.

21) For which of the following flow, the Bernoulli equation will NOT be valid?
a) If the flow is steady and Compressible
b) If the flow is incompressible and frictional effects are negligible c) If the flow is steady and frictional effects are negligible
d) None of the given option.

22) The continuity equation in Cartesian Coordinates for incompressible, two dimensional flow in the "xz" plane will be given as a) $\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0$
b) $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial w} = 0$ c) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ d) $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

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23) Continuity equation is the mathematical form of conservation of mass applied to a fluid particle in a flow.

24) Which of the following is a Navier-Stokes equation for incompressible flow?

$\rho \frac{D\vec{V}}{Dt} = -\vec{\nabla}P + \rho\vec{g} + \mu\nabla^2\vec{V}$

25) If the process is steady, the mass flow rate entering and leaving a control volume must not be equal. False

26) The Bernoulli equation $\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$ is applicable if the flow is steady, incompressible, and irrotational (Topic 7.8, Pg 80)

27) The dimension of the laplace operator $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$ is a) $1/L$ b) $1/t$
c) $1/T$ d) $1/m$

28) For steady, incompressible, laminar flow of a Newtonian fluid is an infinetly round Pipe so that all of the parallel flow, with axis symmetric velocity field and swirls, then which of the following must be true.
For steady, incompressible, laminar flow of a Newtonian fluid in an infinietly long round pipe so that all of the parallel flows, with
a) $U_r = 0$ b) $\frac{\partial U_\theta}{\partial \theta} = 0$ c) $U_z = 0$ d) $U_\theta = 0$

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29) Which of the following fluid property is responsible for the development of the velocity boundary layer?

a) Viscosity b) Velocity c) Pressure d) Density

ATTH642 Subjective 16 March 2022

1) Water discharge to the atmosphere from the orifice at the bottom of a pressurized tank. Assuming frictionless flow, the discharge rate of water from the tank is estimated as 0.168 m^3/s. Discuss with reasons whether the actual discharge rate will be the same, more or less than the estimated value.

Ans:- Since it is frictionless flow system so the discharge rate will be same as that of the actual discharge rate.

2) Define non dimensional variables? Give an example of non dimensional variable?

Ans:- Quantities that changes or vary in the problem, but have no dimension.

Example:- Angle of rotation, measured in degree or radians which are dimensionless units.

Q3) Write the differential form of Newton's Second law of motion in terms of linear momentum. What does this expression state?

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Ans:- As the Newton's second law of motion is given by $\vec{F} = m\vec{a}$.

Where $\vec{a}$ is the acceleration and defined as Time rate of change of velocity.

$$\vec{a} = \frac{d\vec{v}}{dt}$$

So $$\vec{F} = m\frac{d\vec{v}}{dt}$$

$$\vec{F} = \frac{d}{dt}(m\vec{v})$$

Newton's second law is also decision as "Time rate of change of linear momentum".

Qs) Expand $\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot(\rho\vec{V}) = 0$ in to the Cartesian Coordinates where $\vec{V} = \vec{V} = (u, v, w)$.

Ans:- As we Know that ~~Rate of~~ Rate of change of mass within CV

$$\int_{CV} \frac{\partial \rho}{\partial t} dV = \frac{\partial \rho}{\partial t} dx\,dy\,dz$$

Net mass flow rate into CV.

$$= \left(\rho v - \frac{\partial(\rho v)}{\partial y}\frac{dy}{2}\right)dx\,dz + \left(\rho w - \frac{\partial \rho w}{\partial z}\frac{dz}{2}\right)dx\,dy$$

Also, Net mass flow rate ~~out~~ of CV.

$$\sum_{out} \dot{m} = \left(\rho u + \frac{\partial(\rho u)}{\partial x}\frac{dx}{2}\right)dy\,dz +$$

$$\left(\rho v + \frac{\partial \rho v}{\partial y}\frac{dy}{2}\right)dx\,dz + \left(\rho v + \frac{\partial \rho w}{\partial z}\frac{dz}{2}\right)dx\,dy$$

Qs) Write the Continuity equation for Comprenible flow in cylindrical Coordinates. (Topic 124, Pg 132)

In cylindrical coordinates,

$$\frac{1}{r}\frac{\partial(\rho r v_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho v_\theta)}{\partial \theta} + \frac{\partial(\rho v_z)}{\partial z} = 0$$

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7) The equation of motion $\frac{d^2z}{dt^2} = -g$ describing the derivation of an object falling by gravity through a vacuum result in as $z = z_0 + w_0 t + \frac{1}{2} g t^2$ after integrating and applying the initial conditions. Find the non dimensional form of the given result.

Ans: $\frac{d^2z}{dt^2} = -g$ is equation of motion

Dimensional Result of an equation of motion is

$z = z_0 + w_0 t + \frac{1}{2} g t^2$

Primary dimensions of all Parameters are

$[z] = [L], [t] = [t]$

$[z_0] = [L]$, $[w_0] = [L/t]$

$[g] = [L/t^2]$

Non-dimensional Variable are

$z^* = \frac{z}{z_0}$, $t^* = \frac{w_0 t}{z_0}$

$\Rightarrow z = z^* z_0$ $\Rightarrow t = \frac{t^* z_0}{w_0}$

So eqn of motion becomes

$$\frac{d^2z}{dt^2} = -g$$

$$\frac{d^2(z^* z_0)}{d(\frac{t^* z_0}{w_0})^2} = \frac{w_0^2 d^2 z^*}{z_0 dt^{*2}} = -g$$

$$\frac{w_0^2}{g z_0} \cdot \frac{d^2 z^*}{dt^{*2}} = -1$$

As we know that Froude No $Fr = \frac{w_0}{\sqrt{g z_0}}$

Non dimensionalize eq of motion is

$$\frac{d^2 z^*}{dt^{*2}} = \frac{-1}{Fr^2}$$

Non-dimensionalize result:- $z^* = 1 + t^* - \frac{1}{2Fr^2} t^{*2}$

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8) Consider an unsteady, two dimensional velocity field given as $\vec{V} = (u, v) = (2.5 + 0.8x^2)\hat{i}$ through a $+(1.5 + 2.5\sin(\omega t) - 1.6xy)\hat{j}$, where angular frequency ω is equal to 2π rad/s. verify that this flow can be approximated as incompressible?

Sol:- $\vec{V} = (u, v) = (2.5 + 0.8x^2)\hat{i} + (1.5 + 2.5\sin\omega t - 1.6xy)\hat{j}$

$u = 2.5 + 0.8x^2 \rightarrow (1)$ $\quad \therefore \omega = 2\pi$ rad/s

$v = 1.5 + 2.5\sin(\omega t) + 1.6xy \rightarrow (2)$

$\therefore$ Incompressible flow $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow (3)$

$(1) \Rightarrow \frac{\partial u}{\partial x} = 0.8 \times 2 \cdot x$

$= 1.6x$

$(2) \Rightarrow \frac{\partial v}{\partial y} = -1.6(1)x$

$= -1.6x$

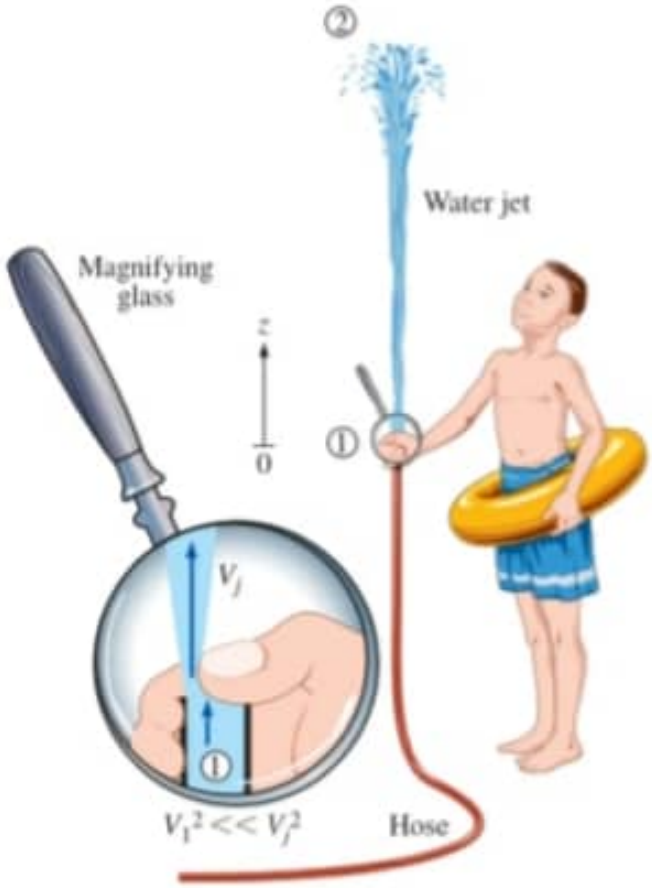
$(3) \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = +1.6x - 1.6x = 0$

$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ Proved

9) Water is following from a house attached to a water main. A child places his thumb to cover most of house outlet, causing a thin jet of high-speed water to emerge attaining the maximum height of 20m if the house is held upward, what could be the Pressure gage in the water main?

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Schematic for Example 5–5. Inset shows a magnified view of the hose outlet region.

Assumptions **1** The flow exiting into the air is steady, incompressible, and irrotational (so that the Bernoulli equation is applicable). **2** The surface tension effects are negligible. **3** The friction between the water and air is negligible. **4** The irreversibilities that occur at the outlet of the hose due to abrupt contraction are not taken into account.

Properties We take the density of water to be 1000 kg/m^3.

Analysis This problem involves the conversion of flow, kinetic, and potential energies to each other without involving any pumps, turbines, and wasteful components with large frictional losses, and thus it is suitable for the use of the Bernoulli equation. The water height will be maximum under the stated assumptions. The velocity inside the hose is relatively low ($V_1^2 << V_j^2$, and thus $V_1 \cong 0$ compared to V_j) and we take the elevation just below the hose outlet as the reference level ($z_1 = 0$). At the top of the water trajectory $V_2 = 0$, and atmospheric pressure pertains. Then the Bernoulli equation along a streamline from 1 to 2 simplifies to

$$\frac{P_1}{\rho g} + \frac{\overbrace{V_1^2}^{\cong 0}}{2g} + \overbrace{z_1}^{0} = \frac{P_2}{\rho g} + \frac{\overbrace{V_2^2}^{0}}{2g} + z_2 \quad \rightarrow \quad \frac{P_1}{\rho g} = \frac{P_{\text{atm}}}{\rho g} + z_2$$

Solving for z_2 and substituting,

$$z_2 = \frac{P_1 - P_{\text{atm}}}{\rho g} = \frac{P_{1,\text{gage}}}{\rho g} = \frac{400 \text{ kPa}}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \left(\frac{1000 \text{ N/m}^2}{1 \text{ kPa}}\right)\left(\frac{1 \text{ kg·m/s}^2}{1 \text{ N}}\right)$$

$$= \mathbf{40.8 \text{ m}}$$

Therefore, the water jet can rise as high as 40.8 m into the sky in this case.

Discussion The result obtained by the Bernoulli equation represents the upper limit and should be interpreted accordingly. It tells us that the water cannot possibly rise more than 40.8 m, and, in all likelihood, the rise will be much less than 40.8 m due to irreversible losses that we neglected.

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MTH642 Quiz 24/3/22

1) 1 atm Pressure is equal to

(a) 105.1 KPa (b) 103.1KPa (c) 102.3KPa (d) 101.3KPa

2) The angular momentum about a Point O is the vector Product of ___ and ___

(a) The Position vector, the force (b) The Position vector, the angular Displacement (c) The Position vector, the momentum

(d) The Position vector, the angular velocity.

3) Non dimensionalized form of Bernoulli equation

$$\frac{P}{P_\infty} + \frac{\rho V^2}{2P_\infty} + \frac{\rho g z}{P_\infty} = \frac{C}{P_\infty}$$

(a) m/t^2L (b) m/L^2t (c) $\{0\}$ (d) $\{1\}$